

**MACHINE LEARNING FUNDAMENTALS - LS2026**  
**SEMINAR: VC DIMENSION**

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**Assignment 1.** What is the VC dimension of the hypothesis class of signed threshold classifiers

$$\mathcal{H} = \{h(x) = \text{sign}(x - \theta) \mid \theta \in \mathbb{R}\} \cup \{h(x) = \text{sign}(\theta - x) \mid \theta \in \mathbb{R}\}?$$

**Solution 1. Answer.** The VC dimension of  $\mathcal{H}$  is

$$\text{VCdim}(\mathcal{H}) = 2.$$

**Proof.**

**1) Lower bound.** We show first that  $\mathcal{H}$  shatters some set of two points. Let  $x_1 < x_2$ . Then all four possible labelings of  $\{x_1, x_2\}$  can be realized:

$$(-, -), \quad (+, +), \quad (-, +), \quad (+, -).$$

Indeed:

- $(+, +)$  is realized by  $h(x) = \text{sign}(x - \theta)$  with  $\theta < x_1$ ,
- $(-, -)$  is realized by  $h(x) = \text{sign}(\theta - x)$  with  $\theta < x_1$ ,
- $(-, +)$  is realized by  $h(x) = \text{sign}(x - \theta)$  with  $x_1 < \theta < x_2$ ,
- $(+, -)$  is realized by  $h(x) = \text{sign}(\theta - x)$  with  $x_1 < \theta < x_2$ .

Therefore,  $\mathcal{H}$  shatters two points, so

$$\text{VCdim}(\mathcal{H}) \geq 2.$$

**2) Upper bound.** Next, we show that no set of three points can be shattered. Let  $x_1 < x_2 < x_3$ . Any hypothesis in  $\mathcal{H}$  is a threshold classifier on the real line, possibly with reversed sign. Hence, along the ordered points  $x_1, x_2, x_3$ , the label can change at most once. In particular, labelings such as

$$(+, -, +) \quad \text{and} \quad (-, +, -)$$

cannot be realized by any  $h \in \mathcal{H}$ . Thus,  $\mathcal{H}$  does not shatter any set of three points, and so

$$\text{VCdim}(\mathcal{H}) < 3.$$

Combining both inequalities in Step 1 and Step 2, we conclude that

$$\text{VCdim}(\mathcal{H}) = 2.$$

**Assignment 2.** Let us consider the class of linear classifiers mapping  $\mathbf{x} \in \mathbb{R}^d$  to  $\{-1, +1\}$ , that is

$$\mathcal{H} = \{h(\mathbf{x}; \mathbf{w}, b) = \text{sign}(\langle \mathbf{w}, \mathbf{x} \rangle + b) \mid (\mathbf{w}, b) \in (\mathbb{R}^d \times \mathbb{R})\}.$$

Show that the VC dimension of  $\mathcal{H}$  is  $d + 1$ .

**Solution 2.** The proof has two steps:

- (1) Show that  $\text{VCdim}(\mathcal{H}) \geq d + 1$ . We construct a set of  $d + 1$  input points and show that they can be shattered.
- (2) Show that  $\text{VCdim}(\mathcal{H}) < d + 2$ . We show that no set of  $d + 2$  input points can be shattered.

**1) Lower bound.**  $\text{VCdim}(\mathcal{H}) \geq d + 1$ . We construct  $d + 1$  points in  $\mathbb{R}^d$  that can be shattered by linear classifiers. Let the first  $d$  points be the standard basis vectors:

$$\mathbf{x}_i = [0, \dots, \underbrace{1}_{i\text{-th coordinate}}, \dots, 0], \quad \forall i \in 1, \dots, d.$$

Let the last point be the origin,

$$\mathbf{x}_{d+1} = \mathbf{0}.$$

Fix an arbitrary sequence of labels  $(y_1, y_2, \dots, y_{d+1}) \in \{-1, +1\}^{d+1}$ . We construct the parameters of the linear classifier as

$$\mathbf{w} = [y_1, y_2, \dots, y_d], \quad b = \frac{1}{2}y_{d+1}.$$

Now verify that the classifier

$$h(\mathbf{x}) = \text{sign}(\langle \mathbf{w}, \mathbf{x} \rangle + b)$$

correctly predicts all  $d + 1$  labels. For  $i \in 1, \dots, d$ ,

$$h(\mathbf{x}_i) = \text{sign}(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) = \text{sign}(y_i + \frac{1}{2}y_{d+1}) = y_i.$$

For the last point,

$$h(\mathbf{x}_{d+1}) = \text{sign}(\langle \mathbf{w}, \mathbf{x}_{d+1} \rangle + b) = \text{sign}(\frac{1}{2}y_{d+1}) = y_{d+1}.$$

Thus, the constructed classifier realizes any labeling of these  $d + 1$  points, proving that the set is shattered. Therefore,

$$\text{VCdim}(\mathcal{H}) \geq d + 1.$$

**2) Upper bound.**  $\text{VCdim}(\mathcal{H}) < d + 2$ . We now show that no set of  $d + 2$  points in  $\mathbb{R}^d$  can be shattered by  $\mathcal{H}$ . Let  $\{\mathbf{x}_1, \dots, \mathbf{x}_{d+2}\} \subset \mathbb{R}^d$  be arbitrary. We lift these points into  $\mathbb{R}^{d+1}$  by appending a constant coordinate:

$$\mathbf{z}_i = [\mathbf{x}_i, 1], \quad \forall i \in 1, \dots, d + 2.$$

We also represent affine classifiers as homogeneous ones:

$$\hat{h}(\mathbf{z}) = \text{sign}(\langle \hat{\mathbf{w}}, \mathbf{z} \rangle), \quad \text{where } \hat{\mathbf{w}} = [\mathbf{w}, b].$$

Clearly,  $h(\mathbf{x}_i) = \hat{h}(\mathbf{z}_i)$  for all  $i$ . Since we have  $d + 2$  points in  $\mathbb{R}^{d+1}$ , they must be linearly dependent. Hence, there exist coefficients  $(a_1, \dots, a_{d+2})$ , not all zero, such that

$$\sum_{i=1}^{d+2} a_i \mathbf{z}_i = \mathbf{0}.$$

Without loss of generality, assume  $a_1 \neq 0$  and express

$$\mathbf{z}_1 = -\sum_{i=2}^{d+2} \frac{a_i}{a_1} \mathbf{z}_i, \quad \text{where } a'_i = -\frac{a_i}{a_1}.$$

Now define labels for the last  $d + 1$  points as

$$y_i = \text{sign}(a'_i), \quad \forall i \in \{2, \dots, d + 2\}.$$

Assume that there exists a classifier  $\hat{h}$  correctly predicting these labels, i.e.

$$\hat{h}(z_i) = y_i \iff \text{sign}(\langle \hat{\mathbf{w}}, z_i \rangle) = \text{sign}(a'_i), \quad \forall i \in 2, \dots, d + 2.$$

Then, for all  $i \geq 2$ , we have  $a'_i \langle \hat{\mathbf{w}}, z_i \rangle \geq 0$ , and for at least one  $i$ , this product is strictly positive. Now, consider the prediction for  $z_1$ :

$$\langle \hat{\mathbf{w}}, z_1 \rangle = \sum_{i=2}^{d+2} a'_i \langle \hat{\mathbf{w}}, z_i \rangle.$$

Because each term  $a'_i \langle \hat{\mathbf{w}}, z_i \rangle \geq 0$ , the sum is non-negative and strictly positive unless all are zero. Hence,

$$\hat{h}(z_1) = \text{sign}(\langle \hat{\mathbf{w}}, z_1 \rangle) = +1.$$

But if we assign the label  $y_1 = -1$ , no classifier can realize this labeling, because  $z_1$  is forced into the positive class. Thus, the set of  $d + 2$  points cannot be shattered by linear classifiers, implying

$$\text{VCdim}(\mathcal{H}) < d + 2.$$

**Assignment 3.** Let  $\mathcal{H} = \{h_i: \mathcal{X} \rightarrow \{-1, +1\} \mid i = 1, \dots, H\}$  be a finite hypothesis class. Show that the VC dimension of  $\mathcal{H}$  is not greater than  $\log_2(H)$ .

**Solution 3.** Assume that  $\mathcal{H}$  shatters a set of  $d$  points,

$$\{x_1, \dots, x_d\} \subseteq \mathcal{X}.$$

By definition of shattering, for every labeling of these  $d$  points by values in  $\{-1, +1\}$ , there exists a hypothesis  $h \in \mathcal{H}$  that realizes this labeling. The number of all possible labelings of  $d$  points is exactly

$$2^d.$$

Hence, if  $\mathcal{H}$  shatters  $d$  points, then  $\mathcal{H}$  must contain at least  $2^d$  distinct hypotheses, because each labeling on  $\{x_1, \dots, x_d\}$  must be realized by some hypothesis. Since  $\mathcal{H}$  contains only  $H$  hypotheses, we must have

$$2^d \leq H.$$

Taking the base-2 logarithm of both sides gives

$$d \leq \log_2 H.$$

Because the VC dimension is the largest integer  $d$  such that some set of  $d$  points is shattered, it follows that

$$\boxed{\text{VCdim}(\mathcal{H}) \leq \log_2 H.}$$

**Assignment 4.** Consider a hypothesis space of classifiers

$$\mathcal{H} = \{h(x; a) = \text{sign}(\sin(ax)) \mid a \in \mathbb{R}\}.$$

That is, each  $h \in \mathcal{H}$  is determined by a single parameter  $a \in \mathbb{R}$  and it maps real valued input  $x \in \mathbb{R}$  to a set of hidden labels  $\{+1, -1\}$  based on the sign of the score  $\sin(xa)$ . Show that the VC dimension of  $\mathcal{H}$  is infinite.

*Hint: Show that for arbitrary set of labels  $\{y^i \in \{+1, -1\} \mid i = 1, \dots, m\}$  the inputs  $\{x^i = 10^{-i} \mid i = 1, \dots, m\}$  can be predicted correctly by  $h(x; a)$  with*

$$a = \pi \left( 1 + \frac{1}{2} \sum_{i=1}^m (1 - y^i) 10^i \right)$$

**Solution 4.** The plan of the proof is to construct points  $(x_1, x_2, \dots, x_m)$  such that for arbitrary labels  $(y_1, y_2, \dots, y_m) \in \{-1, +1\}^m$ , there will be  $a$ , which we also construct, such that all the points will be correctly classified by  $h(x) = \sin(ax)$ , i.e. the set of points is shattered. Thus, we conclude that the VC dimension of  $\mathcal{H}$  is  $\infty$ .

(1) Set the  $m$  points to be

$$x_i = 10^{-i}, \quad i \in \{1, \dots, m\}.$$

and fix an arbitrary sequence of the labels  $(y_1, y_2, \dots, y_m) \in \{-1, +1\}^m$ .

(2) Let the classifier parameter be:

$$a = \pi + \pi \sum_{j=1}^m \hat{y}_j 10^j$$

where

$$\hat{y}_i = \frac{1 - y_i}{2} \in \{0, 1\}.$$

are auxiliary labels.

(3) We will use the identity

$$\sin(\pi(k + t)) = (-1)^k \sin(\pi t),$$

which is valid for every integer  $k \in \mathbb{N}$ .

(4) Let us rewrite the value of the discriminant function at  $x_i$ :

$$\begin{aligned}
 \sin(a x_i) &= \sin \left( \frac{\pi + \pi \sum_{j=1}^m \hat{y}_j 10^j}{10^i} \right) \\
 &= \sin \left( \pi \left[ 10^{-i} + \sum_{j=1}^{i-1} \hat{y}_j 10^{j-i} + \hat{y}_i + \sum_{j=i+1}^m \hat{y}_j 10^{j-i} \right] \right) \\
 &= \sin \left( \underbrace{\pi \sum_{j=i+1}^m \hat{y}_j 10^{j-i}}_{\text{even integer}} + \pi \left( 10^{-i} + \sum_{j=1}^{i-1} \hat{y}_j 10^{j-i} + \hat{y}_i \right) \right) \\
 &= \sin \left( \pi \left( \underbrace{10^{-i} + \sum_{j=1}^{i-1} \hat{y}_j 10^{j-i}}_{\delta \in (0,1)} + \hat{y}_i \right) \right) \\
 &= \sin \left( \pi \left( \delta + \hat{y}_i \right) \right) \\
 &= \begin{cases} \sin(\pi\delta) & \text{if } y_i = +1 \\ \sin(\pi\delta + \pi) & \text{if } y_i = -1 \end{cases} \\
 &= \text{sign}(y_i) .
 \end{aligned}$$